

Designing an Intervention against Occupational Stress Based on Ubiquitous Stress and Context Detection

Mitja Luštrek
Jožef Stefan Institute
Department of Intelligent Systems
Jožef Stefan International
Postgraduate School
Ljubljana, Slovenia
mitja.lustrek@ijs.si

Junoš Lukan
Jožef Stefan Institute
Department of Intelligent Systems
Jožef Stefan International
Postgraduate School
Ljubljana, Slovenia
junos.lukan@ijs.si

Larissa Bolliger
Department of Public Health and
Primary Care
Ghent University
Ghent, Belgium
larissa.bolliger@ugent.be

Emelien Lauwerier
Department of Public Health and
Primary Care
Ghent University
Ghent, Belgium
emelien.lauwerier@ugent.be

Els Clays
Department of Public Health and
Primary Care
Ghent University
Ghent, Belgium
els.clays@ugent.be

ABSTRACT

Stress, particularly occupational, is a major burden for individuals and the society, affecting the health of one worker in four. Digital interventions are a scalable and moderately effective way to tackle it. Just-in-time adaptive interventions (JITIs), which use sensors and artificial intelligence to adapt to the user's needs and context, appear to be a promising way to increase the effectiveness. In this paper we first describe a wearable and mobile data collection as well as stress detection experiments intended to serve as the basis for such an intervention. We then present our design for a JITI based on cognitive behavioural therapy, and highlight a number of open questions applicable to this and similar JITIs.

CCS CONCEPTS

• **Human-centered computing** → Ubiquitous and mobile computing systems and tools; • **Applied computing** → Consumer health.

KEYWORDS

Wearable and mobile data, stress detection, machine learning, just-in-time adaptive intervention, cognitive behavioural therapy

ACM Reference Format:

Mitja Luštrek, Junoš Lukan, Larissa Bolliger, Emelien Lauwerier, and Els Clays. 2023. Designing an Intervention against Occupational Stress Based on Ubiquitous Stress and Context Detection. In *Adjunct Proceedings of the 2023 ACM International Joint Conference on Pervasive and Ubiquitous Computing*

& the 2023 ACM International Symposium on Wearable Computing (UbiComp/ISWC '23 Adjunct), October 08–12, 2023, Cancun, Quintana Roo, Mexico. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3594739.3611326>

1 INTRODUCTION

Stress and the resulting ill health represent a major burden for individuals and the society [20]. Stress directly reduces well-being and contributes to a number of diseases, such as cardiovascular and gastrointestinal, and also has a further, indirect, impact on health through behavioural changes such as unhealthy diet. The leading cause of stress is work, and it is estimated that about one in four workers experience work-related stress causing adverse health effects [9, 32]. Within the EU-15, occupational stress has been estimated to cost the society USD 26.47 billion per year [11]. This makes occupational stress a prime target for study and intervention.

There is lack of health professionals who could assist people suffering from stress and other mental health issues, so digital tools are popular due to their low cost and scalability. The effectiveness of such tools in a clinical setting is well established, e.g., therapist-guided cognitive behavioural therapy (CBT) delivered through the internet was shown to be equally effective for psychiatric disorders as traditional face-to-face treatment [8]. In the general population, digital interventions that include some form of professional guidance have results in line with non-digital treatment for a variety of mental health problems, including stress [16]. Effects were found to be larger in symptomatic populations with elevated stress levels, and for interventions that are CBT-based. CBT is one of the most substantiated approaches to treat (occupational) stress [24]. CBT for stress-related problems generally helps to overcome unhelpful behaviour by correcting false beliefs that are at its core [4]. For example, a person that believes they are worthless or do not work enough, may set unrealistic deadlines and therefore increase burden they put on themselves. CBT helps to challenge these beliefs, in order for new behaviour to arise (e.g., better work-life balance).

Recent reviews show that digital mental health interventions in occupational settings have positive effects, but effects remain modest and do not seem to have increased over the past decade

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

UbiComp/ISWC '23 Adjunct, October 08–12, 2023, Cancun, Quintana Roo, Mexico

© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-0200-6/23/10...\$15.00

<https://doi.org/10.1145/3594739.3611326>

[23, 28]. Effectiveness becomes higher when involving guidance by a human professional, but this impairs the scalability. To increase the effectiveness of digital interventions without involving human professionals, ways to enhance their sophistication are sought. Just-in-time adaptive interventions (JITAI) seem to be a promising approach. They provide the right type and/or amount of support at the right time, i.e., adapted to the needs and context of the user [22]. They use mobile and sensing technology to monitor the user and deliver the intervention, and artificial intelligence (AI) to interpret the user's current context (just in time) and tailor the intervention to their needs (adaptive). While technological capacities for JITAI are increasingly available, research on their development and evaluation lags behind. Most of them have been implemented for behaviour change, e.g., targeting substance misuse and obesity, while applications for (occupational) stress are lacking.

In this paper we propose a JITAI against occupational stress based on CBT. We start by describing extensive data collection with wearable sensors and smartphones aimed at providing a dataset that can be employed for stress detection and behaviour monitoring. We also present preliminary results on stress detection using this dataset. This serves as the foundation of the proposed JITAI. We then proceed with a description of the JITAI itself, outlining the main envisioned features and some open questions regarding the design and evaluation.

2 RELATED WORK

2.1 Stress detection

Stress cannot be directly measured, but can only be observed by either asking the person for their subjective evaluation or measuring its consequences, such as physiological signals. To effectively prevent stress, a monitoring tool would need to be automatic, without requiring user input. There have been multiple attempts to recognize stress from appropriately collected signals.

Typically, the process of building a stress detection system is as follows. First, the data are collected using physiological monitoring devices, smartphones and other ubiquitous devices. Interesting physiological parameters include skin conductivity, heart activity, brain activity, and others [26]. Participants in these data collection studies label their data by responding to questionnaires. These are sometimes composed of simple questions, while other studies use psychologically validated questionnaires [1]. Finally, features are calculated from signals and various machine learning methods are employed to train the predictive models of stress.

Classical machine learning algorithms, such as k-nearest neighbours and support vector machines, have been commonly used for this purpose. Deep learning methods have not been sufficiently explored in the domain of human stress detection [2]. This is probably due to their requirement of large datasets, which are difficult to acquire and especially to label. Additionally, most studies are done on laboratory data, so the methods might not generalize to a real-world use well enough. Another promising aspect to explore is also the long-term reasoning of such models. Since wearables can monitor the user's physiological state continuously and ubiquitously, behavioural patterns might be observed and used for more accurate predictions [25].

2.2 Just-in-time adaptive digital stress interventions

Many solutions exist that use digital interventions, some passive that only remind the user to do an exercise, while others require user input, for example via a chatbot. Stress management tools they use range from educational content (e.g., self-guided training), breathing and mindfulness exercises, offers of a list of positive thoughts and guided exercises for cultivating self-compassion, recording and analysing moods, sleep quality, and enjoyable activities (see Healthy Mind for an example [21]), to using virtual reality to augment these interventions. It is important to note that while these exercises are inspired by CBT or stem from related interventions such as mindfulness, they are not delivering actual therapy [12].

A fully automated JITAI for stress management, which would be based on CBT and adapted for the workplace context, is currently not available. Existing digital CBT interventions for mental health exhibit variability in the degree of professional guidance and automation required [17]. These range from digital CBT supplementing in-person therapy, guided digital CBT that still necessitates some therapeutic feedback, to completely automated CBT without professional guidance; the latter is notably limited, particularly in the realm of stress management. Due to the pervasive nature of smartphones and other smart devices, mobile applications have emerged as a prevalent medium for delivering CBT *in situ*. Compared to internet-delivered solutions, smartphone applications provide real-time engagement in users' everyday contexts, offering an easily accessible, user-friendly, and cost-effective method [16]. These applications also present opportunities for customization to suit individual user needs and daily contexts.

However, most existing solutions do not fully exploit their technological potential, often overlooking smartphone sensor utilization, and rarely incorporating ecological momentary assessment (EMA), especially in conjunction with physiological monitoring [7, 31]. Some more advanced solutions, such as MoodMission [3], tailor the CBT delivery based on user input but do not employ real-time detection methods. Furthermore, a prevalent drawback of current digital CBT applications is their focus on single or limited aspects, such as symptom monitoring or fostering a specific skill like mindfulness. Finally, even the handful of solutions that offer CBT have not been scientifically evaluated [12].

3 DATASET AND METHODS FOR STRESS DETECTION

3.1 Data collection

The phenomenon of chronic stress, which is extensively studied and known to result in adverse health impacts [6], has a less understood connection with daily stress experiences [19]. The Stress at Work (STRAW) project implemented a longitudinal diary study to explore this relationship [5]. We gathered data through three main methods, each employing a different tool for measurement. Empatica E4 wristbands recorded physiological parameters, while a custom-built Android application, based on the AWARE framework [10], gathered smartphone sensor data and phone usage data continuously. Additionally, participants completed questionnaires on their smartphones over 15 working days. The random deployment

of these EMAs occurred throughout the working day, with an approximate gap of 90 minutes. We also distributed a more in-depth questionnaire in the evening to gather information about the entire day. The EMAs included questions drawn from a range of questionnaires intended to evaluate various aspects of stress and related constructs, such as stress appraisal, negative affect, job demand, and job control. We have previously provided a thorough description of the questionnaire contents and data types in an extensive protocol paper [5].

We gathered data from 56 participants that we recruited from academic institutions in Belgium (29 participants) and Slovenia (26 participants). For $N = 55$ participants, we gathered complete data, which included 26 women and 29 men. These participants, with a mean age of 34.9 years and ranging from 24 years to 63 years, held diverse roles in their institutions, such as PhD students, administrative staff, and tenured professors. Participants followed the study protocol diligently. During their involvement, each participant, on average, responded to more than 96 EMA sessions. The median time difference between two consecutive workday EMA sessions was 93 minutes, slightly more than the planned interval [18].

3.2 Stress detection

As the first step in our preliminary analysis, we followed a classical machine learning approach for detecting stress [1]. After preprocessing, we calculated hand-crafted features. For phone sensor data, we used a modified *Reproducible Analysis Pipeline for Data Streams* (RAPIDS, [33]) library, which calculates behavioural features, whereas for physiological data, we used our in-house developed Python library, *cr-features* [13]. In particular, we also developed a human-voice detection method [14], running directly on the smartphone to retain the privacy of participants and obtain some audio information about their surroundings at the same time.

We selected a target variable from the stress measurements, with some (e.g., negative affect) more difficult to predict the others (e.g., general stressfulness). The data were aggregated on segments of variable length, from daily segments down to 30 minutes preceding the labels. The target variables were then averaged and statistical physiological features first calculated on short segments (from 2 seconds to 4 minutes) and then aggregated. Next, we standardized the data *within* participants, i.e., by subtracting the daily mean and dividing by daily standard deviation. Finally, we used a leave-one-subject-out validation technique and tested various linear (e.g., linear regression), non-linear (e.g., support vector regression) and ensemble machine learning techniques (e.g., ADA boost).

We found that a regression problem was very difficult to tackle, as none of the models performed better than a dummy, constant model. We therefore settled for a classification problem by discretizing the target variables into categories of low and high stress. We tested similar models adapted for classification. This turned out to be more manageable, with the best F1 score of 0.441, where roughly half of the items were correctly predicted, while the rest were confused in a balanced way. It should be noted that most of the results from the literature are solving these types of simplified tasks. Furthermore, these are preliminary results; only the first part of the pipeline (i.e., selecting the target variable and the type of prediction) has been explored, and there has been no hyperparameter tuning.

4 PROPOSED INTERVENTION AGAINST OCCUPATIONAL STRESS

4.1 Design of the intervention

It is essential that a mental health intervention satisfy user needs, since such users are particularly unlikely to do deal with a poorly designed application. Therefore we plan to design it in a participatory manner, by involving the potential users from needsfinding and conceptualisation to prototyping. However, based on CBT theory and best practices, we can already plan the main elements of the application [24]. The key CBT element will be a diary of stress experiences implemented in a mobile application, which will be filled semi-automatically with the help of sensors and AI. This personal diary is crucial as it helps to gain insight into how one thinks, feels and acts in stressful situations. The diary entries will be analysed to guide the user through exercises that will help them identify occupational stressors, and maladaptive patterns of thinking, feeling and responding to the stressors. Cognitive re-structuring exercises will help correct unhelpful thoughts and promote adaptive behaviour [4]. Other exercises will also be offered such as relaxation, emotion regulation and problem solving, which are among the most common components in stress CBT interventions [29]. These will all be delivered through the mobile application.

To make our intervention just-in-time and adaptive (i.e., a JITAI), we will utilise AI in several ways. We will use machine-learning models to detect and understand stress and context in order to semi-automatically fill in diary entries. Detection of stress using models based on those described in Section 3.2 will start an entry. This will make the intervention just-in-time (the JIT in JITAI). The entry will then be tagged with information relevant to the user's stress experience. One piece of such information may be that the user was in a meeting, which we are already trying to detect using features corresponding to human voice and location. Another piece of important information may be the people present. These can in many cases be detected based on their Bluetooth devices – initially the user will be asked for their names, but after some time it will be possible to automatically associate the names with the devices. The user will be able to confirm or change all the automatically provided information in the diary entries. In addition, they will be asked for an unstructured report about the stress experience.

The information in the diary will be analysed to find patterns and support reflection on stress (e.g., excessive worrying linked to meetings with a certain person). The user will be recommended evidence-based cognitive exercises, starting with an observation from the diary (e.g., You report certain thoughts in that situation). This will make the intervention adaptive (the A in JITAI). The user may then be asked follow-up questions tailored to the observation, and finally an exercise will be suggested (e.g., Sum up the evidence for and against these thoughts). The user will also be recommended relaxation and other non-cognitive exercises based on the context (e.g., a breathing exercise as soon as they are in their office after a stressful event). The overall operation of the proposed intervention is shown in Figure 1. Considering the explosive development of large language models, an open question is whether and how to use them to provide better CBT exercises. Even generic models can probably be used to extract structured information from unstructured stress reports, and this information can in turn be used to

refine the exercises. They can probably also be used to generate parts of the text of the exercises.

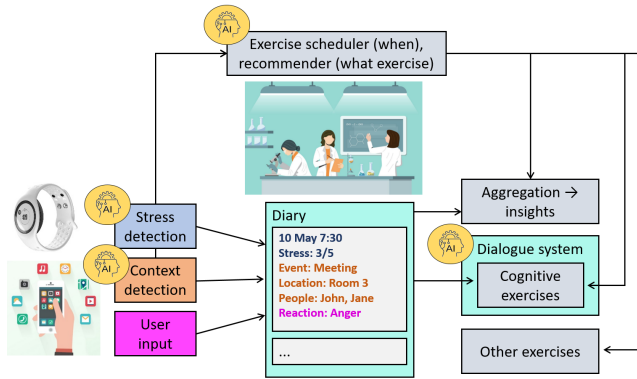


Figure 1: The operation of the proposed just-in-time adaptive intervention against occupations stress.

Yet another use of AI may be to provide personalised recommendations of CBT and other exercises. The choice of the exercises will already depend on the context (e.g., a relaxation exercises after a stressful event and a CBT exercise in the evening at the user’s leisure). But some users in some situations may benefit more from certain types of exercises, so the exercises can be tailored to each user based on their history, and even more flexibly with the help of recommender-system techniques. To accomplish this, we will need an assessment of effectiveness of each exercise. The stress detection models can in principle assess the level of stress before and after an exercise. However, as seen in Section 3.2, this may not be possible to accomplish accurately enough. But then again, it may be possible in a relative manner for each individual. How exactly to do this is another open question, and so is whether personalised exercise recommendations provide enough benefit to be worth the effort.

4.2 Evaluation

When evaluating the proposed JITAI, several important decisions need to be made. The first one concerns what the control condition should be. If one wanted to know whether a digital intervention can reduce stress, the comparison should be against nothing. A step up in difficulty would be a comparison against verbal or written guidance on stress management. However, the literature already reports that digital interventions are effective [23, 28], so our main interest is whether a JITAI (designed as we propose) is better than a “conventional” digital intervention. A natural way to test this is to compare our JITAI to one of such conventional interventions that can be obtained in mobile application stores. Our concern is that these commercial interventions are more mature and aesthetically more pleasing than our research prototype will be, which may give them an advantage even if their mechanisms are simpler and less effective. Therefore we lean towards comparing our JITAI to the same application with the just-in-time and adaptive elements turned off. Another interesting comparison would be to an intervention conducted by a mental health professional, which would presumably show how far the JITAI is from the currently best option. Although it is conceivable that a sufficiently advanced

digital intervention would surpass a human professional due to having more information on the user and being always available.

The second decision concerns what measure of success to use. The obvious option is the difference in the level of stress obtained with an appropriate questionnaire [30] before and after the intervention period, in participants using our JITAI compared to participants in the control condition. Stress could also be evaluated again after a few months to understand long-term effect. However, this option may not be the best match for a JITAI. An alternative is to measure stress experience with an EMA several times a day, which is an approach that has been used to evaluate JITAI in the past [27]. Another option is micro-randomisation [15], which has also been proposed to evaluate JITAI. In this approach, each participant is randomised into the JITAI or control condition each time an action is taken, e.g., an exercise recommended. In our case we would compare JITAI exercises to exercises by the user or recommended periodically at a fixed time, randomising each user each day into the JITAI or control condition. The advantage of this approach is that effect of between-participant differences is eliminated, which can lead to the need for a smaller sample size and/or more reliable results. The disadvantage is that the participants may behave in ways that will invalidate the results, i.e., they may manually start exercises on JITAI days because they will get used to doing so in control days, or they may wait for exercises to start automatically on control days because they will expect them to due to JITAI days. The best evaluation approach is thus still unclear.

5 CONCLUSION

In our work towards an advanced JITAI against occupational stress, we have so far collected a large dataset of wearable and mobile data labelled with different measures of stress. To our knowledge this is the largest dataset in this domain in terms of the measurement points (i.e., the product of the number of people and number of EMA responses for each) [19]. We have also extensively experimented with stress detection. The accuracy is not yet as high as we would like, probably because the dataset was collected entirely in the wild, so we plan to make further refinements to our models.

We are also planning the actual intervention. Our design uses stress detection and the analysis of user context to make the intervention just-in-time. The diary of the user’s stress experiences will be utilised to make the intervention adaptive. The open questions are whether we should attempt to select the best exercises for each user based on how much they reduce stress, and whether and how to incorporate large language models. Regarding the evaluation, we are not yet sure whether to use micro-randomisation and randomise each participant into JITAI or control condition daily, or randomise them once. Whether to measure stress at the beginning and end of the intervention period, daily or multiple times a day also remains an open question.

ACKNOWLEDGMENTS

The STRAW project was funded by the Slovenian Research Agency (N2-0081) and the Research Foundation – Flanders of Belgium (G.0318.18N). The work was also funded by the WideHealth Horizon 2020 project (952279) and Slovenian Research Agency’s core funding (P2-0209).

REFERENCES

- [1] Ane Alberdi, Asier Aztiria, and Adrian Basarab. 2016. Towards an automatic early stress recognition system for office environments based on multimodal measurements. *Journal of Biomedical Informatics* 59 (Feb. 2016), 49–75. <https://doi.org/10.1016/j.jbi.2015.11.007>
- [2] Aamir Arsalan, Syed Muhammad Anwar, and Muhammad Majid. 2022. Mental Stress Detection using Data from Wearable and Non-wearable Sensors: A Review. (2022). <https://doi.org/10.48550/ARXIV.2202.03033> arXiv:2202.03033 [cs.HC]
- [3] David Bakker, Nikolaos Kazantzis, Debra Rickwood, and Nikki Rickard. 2018. A randomized controlled trial of three smartphone apps for enhancing public mental health. *Behaviour Research and Therapy* 109 (Oct. 2018), 75–83. <https://doi.org/10.1016/j.brat.2018.08.003>
- [4] Aaron T Beck. 1967. *Depression. Clinical, Experimental and Theoretical Aspects*. Hoeber, New York.
- [5] Larissa Bolliger, Junoš Lukan, Mitja Luštrek, Dirk De Bacquer, and Els Clays. 2020. Protocol of the Stress at Work (STRAW) Project: How to Disentangle Day-to-Day Occupational Stress among Academics Based on EMA, Physiological Data, and Smartphone Sensor and Usage Data. *International Journal of Environmental Research and Public Health* 17, 23 (Nov. 2020), 8835. <https://doi.org/10.3390/ijerph17238835>
- [6] Daniel J. Brotman, Sherita H. Golden, and Ilan S. Wittstein. 2007. The cardiovascular toll of stress. *The Lancet* 370, 9592 (2007), 1089–1100. [https://doi.org/10.1016/s0140-6736\(07\)61305-1](https://doi.org/10.1016/s0140-6736(07)61305-1)
- [7] Laura Jane Bry, Tommy Chou, Elizabeth Miguel, and Jonathan S. Comer. 2018. Consumer Smartphone Apps Marketed for Child and Adolescent Anxiety: A Systematic Review and Content Analysis. *Behavior Therapy* 49, 2 (March 2018), 249–261. <https://doi.org/10.1016/j.beth.2017.07.008>
- [8] Per Carlbring, Gerhard Andersson, Pim Cuijpers, Heleen Riper, and Erik Hedman-Lagerlöf. 2017. Internet-based vs. face-to-face cognitive behavior therapy for psychiatric and somatic disorders: an updated systematic review and meta-analysis. *Cognitive Behaviour Therapy* 47, 1 (Dec. 2017), 1–18. <https://doi.org/10.1080/16506073.2017.1401115>
- [9] European Foundation For The Improvement Of Living And Working Conditions (Eurofound). 2017. Sixth European working conditions survey—overview report (2017 update). <https://doi.org/10.5255/ukda-sn-8098-4>
- [10] Denzil Ferreira, Vassilis Kostakos, and Anind K. Dey. 2015. AWARE: Mobile Context Instrumentation Framework. *Frontiers in ICT* 2, 6 (2015), 1–9. <https://doi.org/10.3389/fict.2015.00006>
- [11] Juliet Hassard, Kevin R. H. Teoh, Gintare Visockaite, Philip Dewe, and Tom Cox. 2018. The cost of work-related stress to society: A systematic review. *Journal of Occupational Health Psychology* 23, 1 (Jan. 2018), 1–17. <https://doi.org/10.1037/ocp0000069>
- [12] Anna Huguet, Sanjay Rao, Patrick J. McGrath, Lori Wozney, Mike Wheaton, Jill Conrod, and Sharlene Rozario. 2016. A Systematic Review of Cognitive Behavioral Therapy and Behavioral Activation Apps for Depression. *PLOS ONE* 11, 5 (May 2016), e0154248. <https://doi.org/10.1371/journal.pone.0154248>
- [13] Vito Janko, Matjaž Bošić, Junoš Lukan, and Gašper Slapničar. 2021. Library for Feature Calculation in the Context-Recognition Domain. In *Proceedings of the 24th International Multiconference INFORMATION SOCIETY – IS 2021*, Mitja Luštrek, Rok Piltaver, and Matjaž Gams (Eds.), Vol. A. 23–26. https://library.ijs.si/Stacks/Proceedings/InformationSociety/2021/IS2021_Volume_A.pdf
- [14] Marko Katrašnik, Junoš Lukan, Mitja Luštrek, and Vitomir Struc. 2019. Razvoj postopka diarizacije govorcev z algoritmi strojnega učenja. In *Proceedings of the 22nd International Multiconference INFORMATION SOCIETY – IS 2019*, Mitja Luštrek, Rok Piltaver, and Matjaž Gams (Eds.), Vol. A. 57–60. http://library.ijs.si/Stacks/Proceedings/InformationSociety/2019/IS2019_Volume_A.pdf
- [15] Predrag Klasnja, Eric B. Hekler, Saul Shiffman, Audrey Boruvka, Daniel Almirall, Ambuj Tewari, and Susan A. Murphy. 2015. Microrandomized trials: An experimental design for developing just-in-time adaptive interventions. *Health Psychology* 34, Suppl (Dec. 2015), 1220–1228. <https://doi.org/10.1037/hea0000305>
- [16] Jake Linardon, Pim Cuijpers, Per Carlbring, Mariel Messer, and Matthew Fuller-Tyszkiewicz. 2019. The efficacy of app-supported smartphone interventions for mental health problems: a meta-analysis of randomized controlled trials. *World Psychiatry* 18, 3 (Sept. 2019), 325–336. <https://doi.org/10.1002/wps.20673>
- [17] Annemarie I. Luik, Simon D. Kyle, and Colin A. Espie. 2017. Digital Cognitive Behavioral Therapy (dCBT) for Insomnia: a State-of-the-Science Review. *Current Sleep Medicine Reports* 3, 2 (May 2017), 48–56. <https://doi.org/10.1007/s40675-017-0065-4>
- [18] Junoš Lukan, Larissa Bolliger, Els Clays, Oscar Mayora, Venet Osmani, and Mitja Luštrek. 2021. Participants' Experience and Adherence in Repeated Measurement Studies Among Office-Based Workers. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers*. ACM. <https://doi.org/10.1145/3460418.3479367>
- [19] Junoš Lukan, Larissa Bolliger, Nele S. Pauwels, Mitja Luštrek, Deirk De Bacquer, and Els Clays. 2022. Work environment risk factors causing day-to-day stress in occupational settings. *BMC Public Health* 22, 1 (2022). <https://doi.org/10.1186/s12889-021-12354-8>
- [20] David McDaid, Emily Hewlett, and A-La Park. 2017. Understanding effective approaches to promoting mental health and preventing mental illness. *OECD Health Working Papers* 97 (Oct. 2017). <https://doi.org/10.1787/bc364fb2-en>
- [21] Leanne G. Morrison, Charlie Hargood, Veljko Pejovic, Adam W. A. Geraghty, Scott Lloyd, Natalie Goodman, Danus T. Michaelides, Anna Weston, Mirco Musolesi, Mark J. Weal, and Lucy Yardley. 2017. The Effect of Timing and Frequency of Push Notifications on Usage of a Smartphone-Based Stress Management Intervention: An Exploratory Trial. *PLOS ONE* 12, 1 (Jan. 2017), e0169162. <https://doi.org/10.1371/journal.pone.0169162>
- [22] Inbal Nahum-Shani, Shawna N Smith, Bonnie J Spring, Linda M Collins, Katie Witkiewitz, Ambuj Tewari, and Susan A Murphy. 2017. Just-in-Time Adaptive Interventions (JITIs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support. *Annals of Behavioral Medicine* 52, 6 (Dec. 2017), 446–462. <https://doi.org/10.1007/s12160-016-9830-8>
- [23] Elena A Phillips, Vladimir S Gordeev, and Jonas Schreyögg. 2019. Effectiveness of occupational e-mental health interventions: a systematic review and meta-analysis of randomized controlled trials. *Scandinavian Journal of Work, Environment & Health* 45, 6 (June 2019), 560–576. <https://doi.org/10.5271/sjweh.3839>
- [24] Katherine M. Richardson and Hannah R. Rothstein. 2008. Effects of occupational stress management intervention programs: A meta-analysis. *Journal of Occupational Health Psychology* 13, 1 (2008), 69–93. <https://doi.org/10.1037/1076-8998.13.1.69>
- [25] Philip Schmidt, Attila Reiss, Robert Duerichen, and Kristof Van Laerhoven. 2018. Wearable affect and stress recognition: A review. (2018). <https://doi.org/10.48550/ARXIV.1811.08854> arXiv:1811.08854 [cs.HC]
- [26] Nandita Sharma and Tom Gedeon. 2012. Objective measures, sensors and computational techniques for stress recognition and classification: A survey. *Computer Methods and Programs in Biomedicine* 108, 3 (Dec. 2012), 1287–1301. <https://doi.org/10.1016/j.cmpb.2012.07.003>
- [27] Smyth and Heron. 2016. Is providing mobile interventions "just-in-time" helpful? an experimental proof of concept study of just-in-time intervention for stress management. In *2016 IEEE Wireless Health (WH)*. IEEE. <https://doi.org/10.1109/wh.2016.7764561>
- [28] Elizabeth Stratton, Amit Lampit, Isabella Choi, Hanna Malmberg Gavelin, Melissa Aji, Jennifer Taylor, Rafael A. Calvo, Samuel B. Harvey, and Nick Glozier. 2022. Trends in Effectiveness of Organizational eHealth Interventions in Addressing Employee Mental Health: Systematic Review and Meta-analysis. *Journal of Medical Internet Research* 24, 9 (Sept. 2022), e37776. <https://doi.org/10.2196/37776>
- [29] Frank Svärdman, Douglas Sjöwall, and Elin Lindsäter. 2022. Internet-delivered cognitive behavioral interventions to reduce elevated stress: A systematic review and meta-analysis. *Internet Interventions* 29 (Sept. 2022), 100553. <https://doi.org/10.1016/j.invent.2022.100553>
- [30] Jennifer Sweetman, Christina M van der Feltz-Cornelis, Iman Elfeddali, and Edwin de Beurs. 2022. Validation of the work stress screener (WOSS-13) and resilience at work scale (ReWoS-24). *Journal of Psychosomatic Research* 160 (Sept. 2022), 110989. <https://doi.org/10.1016/j.jpsychores.2022.110989>
- [31] Marjolein R. Thunnissen, Marije aan het Rot, Barbara J. van den Hoofdakker, and Maaike H. Nauta. 2021. Youth Psychopathology in Daily Life: Systematically Reviewed Characteristics and Potentials of Ecological Momentary Assessment Applications. *Child Psychiatry & Human Development* 53, 6 (June 2021), 1129–1147. <https://doi.org/10.1007/s10578-021-01177-8>
- [32] O Vargas, J Flintrop, J Hassard, X Irastorza, M Milczarek, JM Miller, A Parent-Thirion, G Van Houten, M Vartiainen, Eurofound, and EU-OSH. 2014. *Psychosocial risks in Europe: Prevalence and strategies for prevention*. Publications Office of the European Union, Luxembourg. <https://doi.org/10.2806/70971>
- [33] Julio Vega, Meng Li, Kwesi Aguilera, Nikunj Goel, Echhit Joshi, Kirtiraj Khandekar, Krina C. Durica, Abhineeth R. Kunta, and Carissa A. Low. 2021. Reproducible Analysis Pipeline for Data Streams. *Frontiers in Digital Health* 3 (2021). <https://doi.org/10.3389/fdgh.2021.769823>